

MLPR Lab-14

This assignment is to classify the Indian currency notes by the use of Transfer Learning. RESNET50 pretrained model will be used for feature extraction through the deep network's layers. Subsequently, SVM will be used for classification.

Think about it for a moment! How wonderful it is that a deep neural network trained on millions of images but which has never before seen any Indian currency note can still be used to extract meaningful features. Thus, in this way, Transfer Learning helps us to adapt such deep neural networks for our much smaller datasets. Otherwise, we would not be able to do classification so well on such small datasets (You can also try Transfer Learning by tuning a few layers of the RESNET50 model later for this or other problems as a take-home exercise)

Instructions:

- Submit all the generated outputs, along with the answers in one PDF.
- Approx. program execution time – 1-2 minutes

Step 1: Import libraries

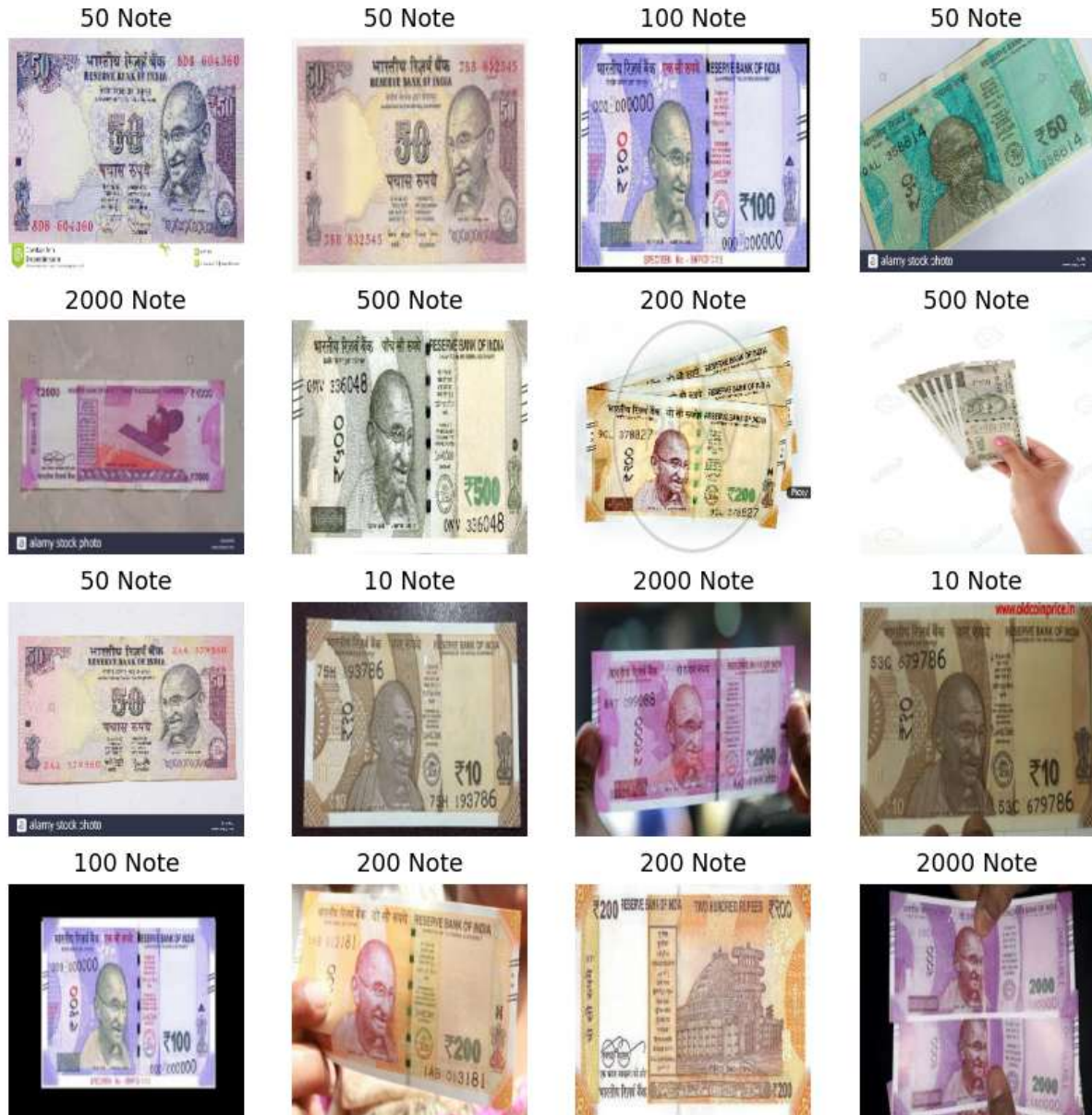
- `import os`
- `import numpy as np`
- `import matplotlib.pyplot as plt`
- `from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay`
- `from tensorflow.keras.preprocessing import image_dataset_from_directory`
- `from tensorflow.keras.applications import ResNet50`
- `from tensorflow.keras import layers, models`
- `from sklearn.svm import SVC`
- `from sklearn.metrics import accuracy_score`

Step 2: Load the dataset- IndianCurrencyNotesDataset/AllImages

- There are seven subfolders inside the folder ALLImages that are seven different values of currencies from note 10 to note 2000.
- You need to extract the data from these subfolders and their labels.
- Do some data augmentation by resizing the images.

Step 3: Split the data with test size 0.2. Do some data augmentation like resizing and rotation only on the train set. Keep batch size 32 and rescale all your images to 224*224 pixels. Display few samples from training dataset with their labels as shown below.

Example Real-world Images from the Dataset

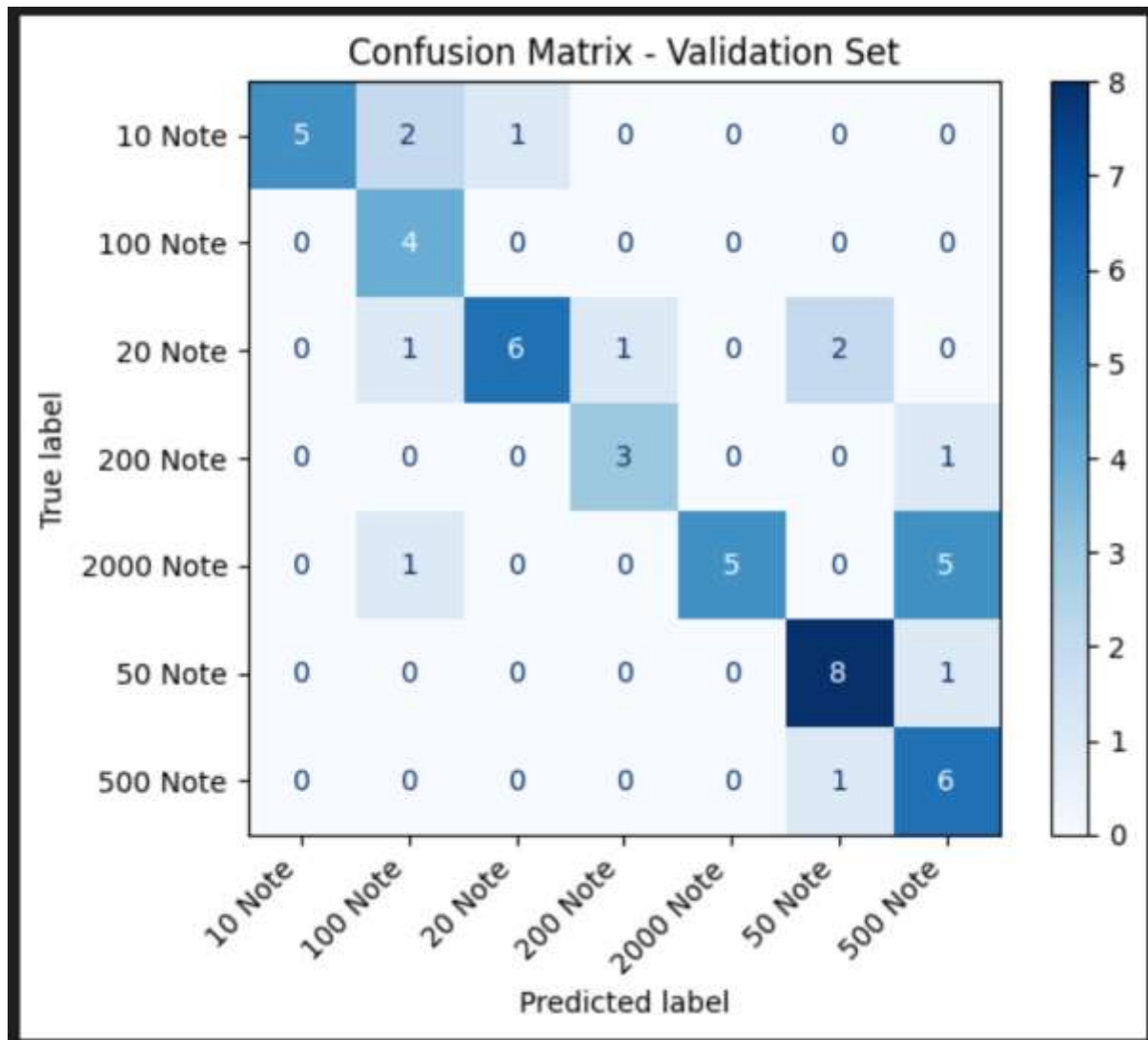


Step 4: Load pretrained ResNet50 model and extract features from the last layer (before the fully-connected layer) using this pretrained network. Store the extracted features and their labels in two separate lists.

Step 5: Now train a SVM classifier over extracted features and labels.

Step 6: Make predictions after you fit the SVC classifier.

Step 7: Calculate accuracy, F1-score, etc. and display confusion matrix as shown below (Values may not exactly match).



Step 8: Display some classified images with their labels.

REPORT QUESTIONS

1. What were the main innovations in ResNet and GoogLeNet?
2. Why do we at times need Transfer Learning?
3. How can up sampling training set help with Transfer Learning?
4. Are there any limitations of using Transfer Learning? List only two.
5. What are the different approaches in Transfer Learning?

DELIVERABLES: Items to be uploaded on LMS-

Main code file along with the complete code, all the generated outputs, and the report answers in one file in PDF format.